

3	(a)	By conservation of energy, Gain in kinetic energy = Loss in electric potential energy $\frac{1}{2}mv^2 - 0 = qV$ $v = \sqrt{\frac{2qV}{m}}$	1
	(b)	The moving ion, upon entering the magnetic field, will experience a constant magnetic force perpendicular to the direction of velocity. This constant magnetic force provides the centripetal acceleration, causes the velocity to change its direction but not the speed (its magnitude). As a result, the ion moves in a uniform circular motion in the magnetic field.	1 1
	(c)	Magnetic force on ion provides the centripetal force. $Bqv = mv^2/r \rightarrow r = mv/Bq = \sqrt{\frac{2mV}{B^2q}}$ $r_{14} = \sqrt{\frac{2 \times 14 \times 1.66 \times 10^{-27} \times 4000}{0.50^2 (1.60 \times 10^{-19})}} = 0.068176 \text{ m}$ $r_{12} = \sqrt{\frac{2 \times 12 \times 1.66 \times 10^{-27} \times 4000}{0.50^2 (1.60 \times 10^{-19})}} = 0.063119 \text{ m}$ Hence $\Delta x = 2(0.068176 - 0.063119) = 0.0101 \text{ m}$	1 1 1
	(d)	If an electron were to be introduced into the magnetic field, it would be deflected to the left because electron is negatively-charged. The electron will move towards the left in a semicircle in an anticlockwise manner. In addition, the diameter of its semi-circular path would be much smaller since the mass of electrons is only much smaller at $9.11 \times 10^{-31} \text{ kg}$.	1 1